

Honeyfish - a high resolution gesture recognition system based on capacitive proximity sensing

Tobias Grosse-Puppendahl¹ and Andreas Braun²

¹TU Darmstadt, Karolinenplatz 5, 64289 Darmstadt, Germany

²Fraunhofer IGD, Fraunhoferstr. 5, 64283 Darmstadt, Germany

`t_groae@rbg.informatik.tu-darmstadt.de, andreas.braun@igd.fraunhofer.de`

Abstract. The recognition of gestures in free space using sensors that determine the proximity of a body mass based on electric field variance is a challenging research topic. Arrays of such capacitive proximity sensors allow the creation of novel user interaction systems based on recognizing presence and position of body parts and inferring performed gestures in three dimensions from that information. These systems may be incorporated as unobtrusive remote control in home automation scenarios or automotive applications, for example as a smart car-dashboard. Present systems that use time-division multiplexing have limitations in their temporal and spatial resolution. In this paper we present the Honeyfish - a novel capacitive proximity sensing system that uses a combination of frequency division and time division multiplex, improving both temporal and spatial resolution. With this affordable system we are able to detect fast multi-hand gestures in three dimensions above large surface areas.

Keywords: Gesture recognition, capacitive proximity sensing

1 Introduction & Related Work

A growing number of devices in one's environment is making interaction with computer-systems more and more complex. Technologies like multi-touch enabled smartphones and tablet pcs show that there is a great demand on systems with low interaction barriers. However, these devices have only pervaded a very small part of our environment - there still exists a broad variety of devices with complex interaction patterns, such as car dashboards or remote controls.

It is foreseeable that the interaction with low-level technical devices is shifted to a more natural way in the future. Therefore, new interaction modalities have to be considered. Beside multi-touch and camera-based interaction systems, the research area of capacitive proximity sensing for human-computer-interaction rises up. Employing this technology, it is possible to realize new kinds of interfaces that are able to determine multi-hand gestures and body movements in free space. In contrast to camera-based methods, capacitive proximity sensing faces the great advantage of being robust against changing lighting conditions and visual occlusion. Sensed data can be processed with computationally cheap

algorithms. However, the drawbacks are a limited resolution and error-proneness in environments with changing conditions affecting electric fields.

Capacitive proximity sensing is based on the principle that electric fields change their properties when a ground-coupled object is moved into them. Three different measurement modes for capacitive proximity sensing can be identified, that are shown in Figure 1. In transmit mode, the distance to a human body part can be determined by coupling the body to a changing electric potential and receiving the transmitted signal using a grounded plate. Shunt mode, which is employed in the presented hardware platform, is based on the principle that a grounded body part affects the electric field between a transmitter and a receiver electrode. Using loading mode, one is able to measure the displacement current from a transmitter electrode to a grounded body part, in order to determine the distance.[6]

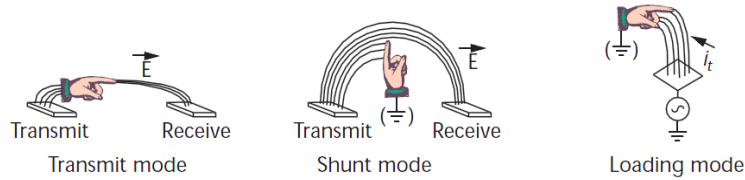


Fig. 1. The three different measurement modes for capacitive proximity sensing [4]

It is necessary to define the word sensor in the context of capacitive proximity sensing. In transmit mode, a sensor can be defined as a receiver, measuring the incoming displacement current. Similar to that, in loading mode, the displacement current is measured by the transmitter, which is considered to be a sensor. In shunt mode the definition of a sensor is not intuitively apparent. As each transmitter-receiver combination may deliver a unique measurement result, a sensor is defined as a combination between a transmitter and a receiver.

The research area of capacitive proximity sensing was established in 1920 by Leon Theremin, with his invention of a music instrument, that he named after himself. With this invention, he did not only lay the foundation to electronic music, but also to capacitive proximity sensing. Figure 2 shows Leon Theremin playing his instrument by moving his hands close to two antennas that are used to control the pitch and volume.[2]

In the early 1990s Smith et. al. started to develop devices and models for human-computer-interaction with capacitive proximity sensors. Within the next ten years, his group developed the first generic approach for recognizing objects, like hands, with electric fields. The author provided the first mathematical and electro-statistical models with high precision [6]. In 2006, Wimmer et. al presented the first gesture recognition system for single hands, working with four capacitive proximity sensors [7] using shunt mode. The system is able to recognize pick and drop gestures in front of a screen. In the following year, Wimmer et. al. presented a generic hardware platform for loading-mode measurements,

called CapToolKit [8]. With CapToolKit, the authors realized a broad variety of applications, reaching from a table tracking hands and movements to smart furniture. An example application using this platform is shown in Figure 3.



Fig. 2. Leon Theremin playing his electronic instrument [2]

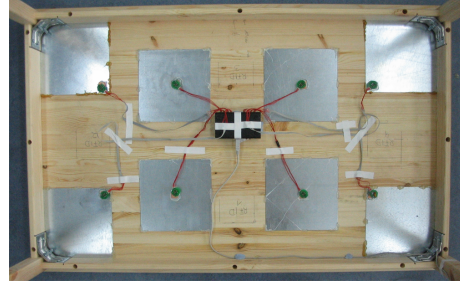


Fig. 3. The Captable is able to recognize the position of hands above its surface [8]

In a recent work, we integrated capacitive proximity sensors in Ambient-Assisted-Living (AAL) environments [1]. Therefore, we focused on explicit human-computer-interaction using a gesture recognition device. Moreover, non-obtrusive AAL-technologies like sensing beds and smart floors, able to contribute to fall detection and people localization, were presented. We have also investigated an implicit interaction modality by classifying human postures on furniture with capacitive proximity sensors [3].

2 Honeyfish concept

2.1 Measurement modes

Three different modes in capacitive proximity sensing can be distinguished, as mentioned in the introduction. Transmit mode measurements were realized in few applications and leak of several weaknesses [5]. Using transmit mode, it is necessary that an object, or a person, is coupled to a changing electric potential. This is not always applicable, as there are two major problems with this measurement mode in terms of human interaction. Firstly, a person is not always willing to act as a human antenna, even though no objective health risks can be identified. And secondly in many scenarios it is not possible to link the person to a transmitter since the interaction is required to be spontaneous and shall not provide too high interaction barriers.

Applying loading mode, one can avoid the user acceptance problem, as well as avoiding coupling the user to an electric potential. In loading mode, a user acts as a grounded object in the environment. Thus, when a user's body part approaches an electrode with changing electric potential, a displacement current from the electrode to the body part can be measured. It is also possible to

measure the resulting capacitance by creating a resonant circuit and measuring the resonance frequency, as employed in [8]. Using N electrodes, one is able to obtain N distinct measurements per cycle. Measurements can be made in parallel, without switching off any electrodes. Thus, this mode is very applicable for many applications, including gesture recognition[1] [7] [8].

In shunt mode, a receiver electrode measures a displacement current from a transmitter electrode. When a human body part enters the electric field, the displacement current becomes smaller, as well as the capacitance between the transmitter and receiver decreases. Due to separate transmitter and receiver electrodes shunt mode offers the possibility to perform parallel measurements. The number of possible measurements is $N = R \cdot T$, increasing with the number of dedicated receivers R and the number of dedicated transmitters T . Employing a number of S transceivers makes $N = 0.5 \cdot (S) \cdot (S - 1)$ distinct measurements possible [5].

Considering the example of an application requiring at least nine distinct sensor measurements loading mode would require nine electrodes and sensing circuits to meet this requirement. In shunt mode, one is able to obtain the same number of measurements with six electrodes (three dedicated transmitters, three dedicated receivers). Thus, the number of possible shunt mode measurements outperforms loading mode measurements when more than one receiver and more than two transmitters are employed. In conclusion loading mode can be performed with less complexity, but shunt mode can provide a higher number of measurements with less electrodes and peripherals.

2.2 Multiple-access schemes

For shunt mode measurements, it is necessary to control the access to the communication channel. A shared access can be realized in the time, frequency, code and spatial domain. Considering gesture recognition the electrodes have to be densely placed in space, therefore only the first three options can be considered.

In capacitive proximity sensing typical transmitter voltages of up to 80 V are used with higher output voltages increasing signal-to-noise ratio and detection range. The applied frequencies are usually within a range of 1 to 100 KHz, whereas working with smaller frequencies is computationally less expensive without analog demodulation circuits employed. [4, 8].

Considering gesture recognition approximately 20 measurements from all sensors are required per second. If the sensor surface is approximately 50 cm of length or width a hand swiping over that surface will cross the area in about 200 ms for typical movement speeds. In order to gather meaningful measurements approximately nine sensors are required. Therefore a multiplexing approach allowing an update rate of at least 20 Hz for all sensors is required for gesture recognition. Furthermore, the multiplexing approach must operate with a low amount of filtering to avoid latency in gesture recognition. Considering the characteristic voltages and frequencies used for capacitive proximity sensing, the identified update rate cannot be achieved with a high number of sequentially enabled transmitters. A frequency-division multiple-access can overcome

this problem by providing a dedicated frequency channel for a group of sequentially enabled transmitters.

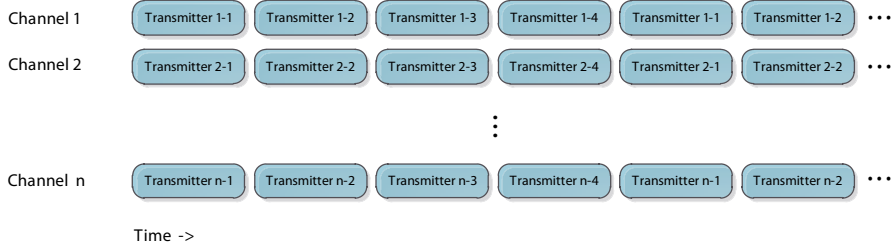


Fig. 4. A combination of FDMA and TDMA provides a good basis for gesture recognition

Using a sampling rate of 50 KHz, a 256-point fast Fourier transform can be performed each 12 ms. Even a frequency channel with 4 time-divisioned transmitters with a measurement cycle of 48 ms, would fulfil the constraint of having at least 20 measurement cycles per transmitter per second. As frequency channels can be used in parallel, additional transmitters operating on a different channel do not have a decreasing effect on the number of measurements. The combination of frequency division multiple-access (FDMA) with time division multiple-access (TDMA) is illustrated in Figure 4.

Shunt mode with a combination of FDMA and TDMA is a suitable solution to achieve a good spatial and temporal system performance. Moreover increasing the number of transmitters and receivers effect the number of measurements and therefore accuracy of the gesture recognition. On this basis the Honeyfish sensor array was developed that will be presented in the next chapter.

2.3 Object recognition

The goal of object recognition is to set up a model that can relate sensor readings to the state of one or more objects in space. As a typical example, one would infer the positions of one or more hands. In capacitive proximity sensing, Smith et. al. introduced the terminology of iso-signal shells, that represent surfaces of constant measurements for a particular object around a sensor [4]. For example, a hand that is posed beside a sensor may produce the same measurement like a hand that is located above the sensor. This ambiguity must always be considered when dealing with such systems.

Combining the measurements of many sensors, one can derive a hand position through the intersections of iso-signal shells, as shown in Figure 5. The Figure depicts three sensors with iso-signal-shells, modelled as ellipsoids. The intersection of these iso-signal shells can be regarded as the object position.

When all sensors are located in a plane, the location is ambiguous and can either be above or beneath that plane. In order to reliably recognize multiple

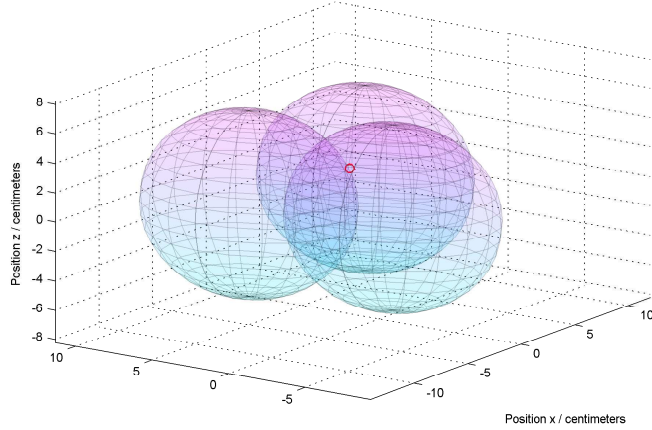


Fig. 5. Iso-signal shells and their intersection above the sensing plane

objects, more sophisticated models are required, which will be part of our future work.

3 Prototype

To prove the feasibility of the presented signal processing approach, a sensor array for gesture recognition was designed. The hardware platform performs shunt mode measurements using a combination of FDMA and TDMA for shared channel access. The array is composed of up to 16 Honeyfish-Boards, each having four transmitters and one receiver. A microcontroller is responsible for controlling the Honeyfish boards' transmitters and decoding the signals of all receivers. Moreover, the microcontroller adjusts the operating frequency of all Honeyfish boards for frequency-division multiplexing. A Honeyfish board has two main tasks. It generates a sine wave with a variable frequency and leads the signal to the four transmitters. Furthermore, it has one receiver circuit, that is employed to amplify received signals from the transmitters, also from different Honeyfish boards.

In Figure 6, a basic block diagram of the transmitter circuit is given. This part of a Honeyfish board contains a sine generator that is adjusted by a pulse-width-modulation (PWM) provided by a microcontroller. Using the PWM, the frequency can be modified within a range of 1 KHz to 12 KHz. The analog switch forwards the sine output to the enabled transmitters. The transmitter's output voltage level lies at approximately 4 V DC.

The block diagram for the broadband receiver circuit is depicted in Figure 3. The receiver must be able to receive signals in the frequency range from 1 to 12 KHz. In the first processing step, a low-noise preamplifier amplifies the received signal. After voltage shift and low-pass filtering, the signal can be sampled by a microcontroller. Sampling is performed with a rate of 50 KHz, applying an

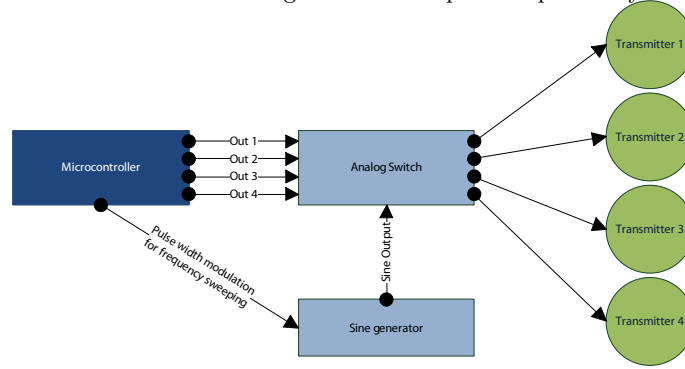


Fig. 6. Block diagram of the transmitter

FFT for the reconstruction of the displacement current within each frequency channel.

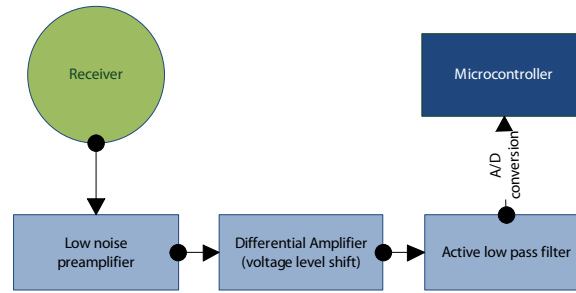


Fig. 7. Block diagram of the receiver circuit

A prototype was developed, employing a Honeyfish sensor array. The prototype is supposed to recognize gestures, performed by either one or two hands. Using two Honeyfish boards with four transmitters and one receiver each, one can yield a number of 16 sensor readings that is sufficient for recognizing up to two hands. The prototype, a box with a length of 50 cm and a width of 30 cm, is shown in Figure 8. In this setup, the transmitters of each Honeyfish board are placed separately on the sides of the box.

Transmitters that are far away from a receiver have a worse signal-to-noise ratio than transmitters that are close to receivers. Assuming that the signal strength from transmitter 1-4 to receiver 2 is 100 %, the signal strength from transmitter 1-2 to receiver 2 lies at 60 %. The resulting signal strength from the farthest transmitter 1-1 to receiver 1 is only 10 %.

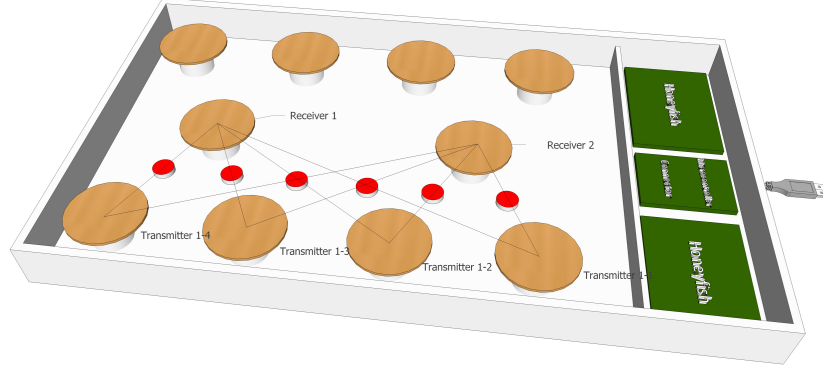


Fig. 8. Virtual sensors in the Honeyfish prototype

4 Discussion

4.1 Evaluation

The resolution of a single sensor within the built prototype was evaluated. Regarding a gaussian noise model, the resolution expresses that 68.3 % of all readings lie within the depicted range. It was determined by taking a series of measurements and calculate average and standard deviation. Therefore, a grounded aluminium tube with a diameter of 8 cm and a length of 48 cm, acting as a surrogate arm, was moved towards the center of a transmitter-receiver pair. The electrodes used in this setup have a diameter of 4 cm and are spaced with a distance of 10.5 cm. They were connected to the Honeyfish board by 20 cm long co-axial cables. The test setup is similar to [8].

A series of succeeding measurements was recorded for varying aluminium tube distances. Figure 9 depicts the average sensor readings and standard deviations that can be obtained for a series of 200 succeeding measurements for each measured tube distance.

It is noteworthy that the resolution in real applications can be increased when combining the measurements of many sensors. Moreover, it is considerable to apply filtering, depending on the application's requirements to latency and update rate. In order to obtain a low noise level, it is necessary reduce the length of the electrodes' cables to a minimum. By using larger electrodes, one can extend the detection range and resolution significantly.

4.2 Conclusion & Future work

On the basis of capacitive proximity sensors, we created a hardware platform for gesture recognition providing a high spatial and temporal resolution. Shunt mode is particularly suitable for gesture recognition as it allows a high number of measurements while having a small number of electrodes and peripherals.

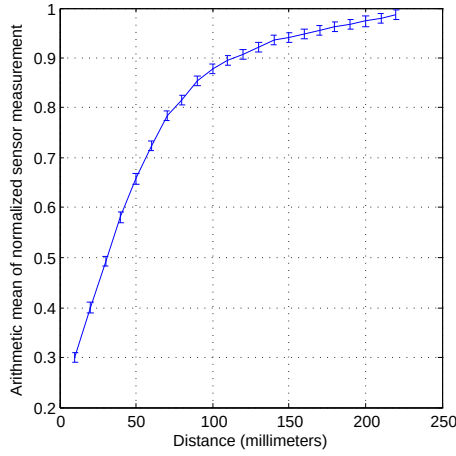


Fig. 9. Averaged normalized sensor readings for various distances of the test object to a receiver-transmitter combination

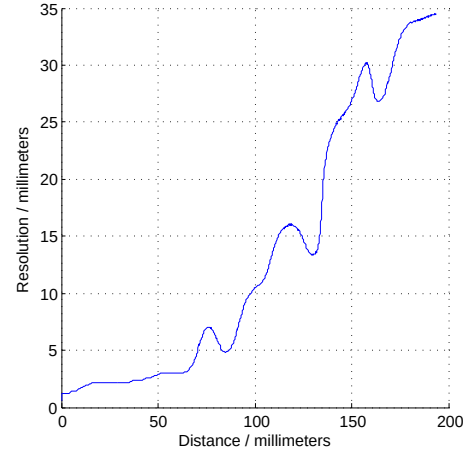


Fig. 10. The resolution of an unfiltered series of sensor readings depending on the measurement noise

The presented multiplexing approach is practicable for an efficient shared access on the communication channel with many transmitters, while maintaining the required temporal resolution for gesture recognition.

Beside the hardware platforms, we will focus on models for gesture recognition. When dealing with continuous gestures, for example controlling a cursor, the position of a body-part must be determined precisely. Especially in the area of continuous gestures, approaches for object tracking are required. Discrete gestures, such as a simple command, can be recognized directly from the sensor readings in a computationally cheap manner, without the need for a sophisticated object recognition model. The reliable recognition of continuous and discrete gestures will be subject of our future work.

In-the-air gesture recognition employing capacitive proximity sensing offers great opportunities for natural interaction within the home-automation and automotive domain. These systems can be embedded into car dashboards, steering wheels or smart tables to realize an easy and comprehensive interaction modality. Especially the low cost, practical applicability and small energy consumption excell those systems for the presented application scenarios.

References

1. Braun, A., Hamisu, P.: Using the human body field as a medium for natural interaction. of the 2nd International Conference on p. 50 (2009), <http://portal.acm.org/citation.cfm?id=1579114.1579164>
2. Glinisky, A.: Theremin: ether music and espionage. Music in American life, University of Illinois Press (2000), <http://books.google.com.au/books?id=6DH1QJcMpBQC>

3. Große-Puppenthal, T.A., Marinc, A., Braun, A.: Classification of user postures with capacitive proximity sensors in aal-environments. In: Keyson, D.V., Maher, M.L., Streitz, N., Cheok, A.D., Augusto, J.C., Wichert, R., Englebienne, G., Aghajan, H.K., Kröse, B.J.A. (eds.) *AmI. Lecture Notes in Computer Science*, vol. 7040, pp. 314–323. Springer (2011)
4. Smith, J.R.: Field mice: extracting hand geometry from electric field measurements. *IBM Syst. J.* 35, 587–608 (September 1996), <http://dx.doi.org/10.1147/sj.353.0587>
5. Smith, J., White, T., Dodge, C., Paradiso, J., Gershenfeld, N., Allport, D.: Electric field sensing for graphical interfaces. *IEEE Comput. Graph. Appl.* 18, 54–60 (May 1998), <http://dx.doi.org/10.1109/38.674972>
6. Smith, J.R.: Electric field imaging. Tech. rep. (1999)
7. Wimmer, R., Holleis, P., Kranz, M., Schmidt, A.: Thracker - using capacitive sensing for gesture recognition. In: *ICDCS Workshops*. p. 64. IEEE Computer Society (2006)
8. Wimmer, R., Kranz, M., Boring, S., Schmidt, A.: A capacitive sensing toolkit for pervasive activity detection and recognition. In: *PerCom*. pp. 171–180. IEEE Computer Society (2007)